IOT BASED SMART ENERGY MONITORING SYSTEM AND LOAD CONTROL

A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRICAL AND ELECTRONICS ENGINEERING

KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

(Autonomous Institution)

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APRIL 2026

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ACKNOWLEDGMENT

We express our sincere thanks and profound gratitude to **Dr. K. P. Ramasamy**, Chairman, and **Ms. Gayathri Anandakrishnan**, Secretary, KPR Group of Institutions, for providing an excellent academic environment and for their visionary leadership and unwavering support for research and innovation.

We are deeply grateful to **Dr. A. M. Natarajan**, Chief Executive; **Dr. R. Devi Priya**, Principal; and **Dr. G. Anusha**, Vice Principal, KPR Institute of Engineering and Technology, for their esteemed guidance and for fostering a culture of academic excellence.

Our sincere gratitude to the coordinator; **Dr. A.Mohamed Ibrahim** Embedded System Design Project mentor and coordinator for his structured oversight and timely advice.

We are especially thankful to **Dr. K. Mohana Sundaram**, Professor and Head, Department of Electrical and Electronics Engineering, for his support.

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ABSTRACT

The IoT-based Smart Energy Monitoring System is an intelligent and efficient solution developed to monitor, control, and optimize electrical energy consumption using Internet of Things (IoT) technology. In recent years, the rapid increase in energy demand and the need for efficient utilization have led to the development of smart monitoring systems that provide real-time insights into energy usage.

This system uses voltage and current sensors to measure electrical parameters from connected loads continuously. The sensed analog signals are converted into digital values and processed by a microcontroller such as ESP8266 or ESP32. The system calculates key electrical parameters including voltage, current, power, and total energy consumption. These values are essential for analyzing the performance and efficiency of electrical devices.

The processed data is transmitted to a cloud server using WiFi communication, enabling remote monitoring through mobile applications or web dashboards. Users can access real-time data, view historical usage patterns, and analyze energy consumption trends from anywhere. This helps in identifying high energy-consuming devices and taking necessary actions to reduce power wastage.

Additionally, the system can be enhanced with smart features such as automatic alerts for abnormal energy usage, overload protection, and remote switching of electrical appliances. These features improve safety, reliability, and user convenience. The system is cost-effective, easy to install, and scalable, making it suitable for residential, commercial, and industrial applications.

By promoting energy awareness and efficient utilization, the IoT-based Smart Energy Monitoring System contributes to reducing electricity costs and supports sustainable energy management practices. It plays a vital role in the development of smart homes and smart grid technologies, ensuring better control over energy resources and minimizing environmental impact.

CHAPTER 1
INTRODUCTION

Energy is one of the most important resources in modern society, and its efficient utilization has become a major concern due to increasing demand and limited availability. Traditional energy monitoring systems are mostly manual and do not provide real-time information, which often leads to energy wastage, higher electricity bills, and lack of proper energy management.

With the advancement of Internet of Things (IoT) technology, it has become possible to develop smart systems that can monitor and control electrical energy consumption remotely. The IoT-based Smart Energy Monitoring System is designed to provide a modern and efficient solution for tracking energy usage in real time. This system uses sensors to measure electrical parameters such as voltage and current, and a microcontroller to process the data and calculate power consumption.

The collected data is transmitted to a cloud platform using WiFi technology, allowing users to monitor their energy usage through mobile phones or web applications from anywhere. This real-time monitoring helps users to understand their consumption patterns, identify energy wastage, and take necessary actions to improve efficiency.

The system uses wireless communication technologies like WiFi to send data to cloud platforms, where it can be stored and accessed remotely. Users can monitor their electricity usage through mobile phones or web applications from anywhere at any time. This helps in understanding energy consumption patterns and taking necessary actions to reduce unnecessary usage.
Moreover, the system can be expanded to include advanced features such as automatic alerts, overload protection, and remote control of electrical appliances. These features enhance safety, convenience, and energy conservation.

The system is cost-effective, user-friendly, and suitable for applications in homes, industries, and commercial buildings.Thus, the IoT-based Smart Energy Monitoring System plays a vital role in promoting smart energy management, reducing power consumption, and supporting sustainable development.

CHAPTER 2
LITERATURE SURVEY

The development of smart energy monitoring systems has gained significant attention in recent years due to the increasing demand for efficient energy utilization and the need to reduce power wastage. Traditional energy monitoring methods mainly rely on manual meter readings, which are time-consuming, less accurate, and do not provide real-time information. To overcome these limitations, researchers have introduced IoT-based solutions that enable automated and continuous monitoring of electrical energy consumption.

Many studies have focused on the use of embedded systems combined with IoT technology to design smart energy meters. These systems typically use sensors to measure voltage, current, and power, and microcontrollers to process the data. The processed data is transmitted to cloud platforms through wireless communication technologies such as WiFi, GSM, or Bluetooth. This allows users to access real-time data and monitor their energy consumption remotely.

Some researchers have proposed systems that include advanced functionalities such as load control, fault detection, and overload protection. These features improve the safety and reliability of electrical systems. In addition, several works highlight the importance of integrating energy monitoring systems with smart home automation, where electrical appliances can be controlled automatically based on energy usage patterns.

Recent studies also emphasize the use of data analytics and machine learning techniques to analyze energy consumption data and predict future usage. This helps in efficient energy planning and reduces unnecessary consumption. Moreover, IoT-based monitoring systems are being integrated with renewable energy sources such as solar and wind to create sustainable energy management solutions.

Despite the advantages, researchers have identified certain challenges such as data security, privacy issues, sensor calibration errors, and network reliability. Ensuring accurate measurements and secure data transmission remains a key focus area for further research. Additionally, scalability and cost-effectiveness are important factors for large-scale implementation.In conclusion, the literature shows that IoT-based smart energy monitoring systems provide an effective solution for real-time monitoring, energy conservation, and smart energy management. Continuous improvements in technology are expected to make these systems more efficient, reliable, and widely adopted in the future.

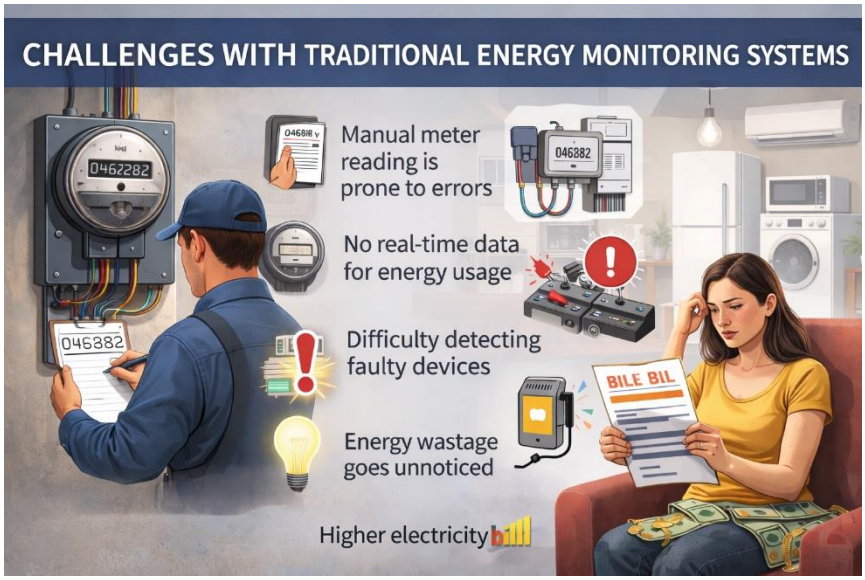
CHAPTER 3

PROBLEM IDENTIFICATION

CHALLENGES WITH TRADITIONAL ENERGY MONITORING SYSTEMS :

Traditional energy monitoring systems, commonly used in homes and industries, face several issues:

- **LACK OF REAL-TIME MONITORING:** Users cannot track electricity consumption instantly, leading to poor energy management.
- **MANUAL METER READING:** Energy usage is recorded manually, which is time-consuming and may lead to human errors.
- **NO REMOTE ACCESS:** Users cannot monitor their energy consumption from remote locations.
- **ENERGY WASTAGE:** Due to lack of awareness, unnecessary power consumption often goes unnoticed.
- **NO FAULT DETECTION:** Traditional systems cannot easily detect issues like overload, power leakage, or abnormal usage.
- **LIMITED DATA ANALYSIS:** No proper system to store and analyze past energy consumption data.
- **HIGHER ELECTRICITY BILLS:** Inefficient monitoring leads to increased power usage and higher costs.
- **LACK OF AUTOMATION:** No automatic control of electrical appliances based on energy usage.



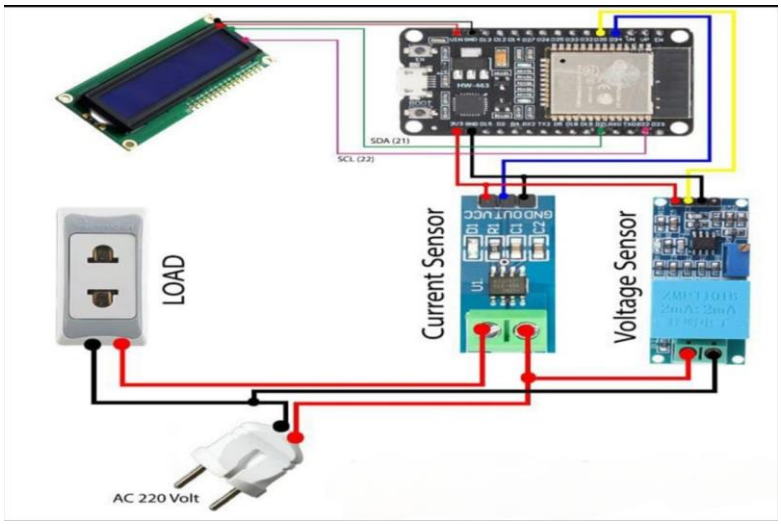
CHAPTER 4
METHODOLOGY

The IoT-based Smart Energy Monitoring System works in a systematic and continuous manner to measure and monitor energy usage. The methodology begins with sensing the electrical parameters from the load. Voltage and current sensors are connected to the circuit, which continuously measure the electrical values and send signals to the microcontroller.

The microcontroller (such as ESP8266 or ESP32) receives these signals and converts them into digital data. It then processes the data to calculate important parameters like voltage, current, power, and total energy consumption. These calculations help in understanding how much electricity is being used by different appliances.

Once the data is processed, it is transmitted through a WiFi module. This allows the user to access the data remotely. The system can display real-time energy usage, which helps users to monitor their consumption regularly. It also helps in identifying unusual or excessive power usage.

In addition, the system can store the collected data for future reference. This stored data can be used to analyze energy consumption patterns over time. Based on this analysis, users can take necessary steps to reduce energy wastage and improve efficiency.The methodology ensures that the entire process—from sensing to monitoring—is automatic, continuous, and user-friendly. It reduces manual effort and provides accurate and real-time information about energy usage.



CHAPTER 5

DESIGN AND FABRICATION

DESIGN OVERVIEW :

The IoT-based Smart Energy Monitoring System is designed to measure and monitor electrical energy consumption in real time. The system consists of sensors, a microcontroller, and a communication module. Voltage and current sensors are used to collect electrical data from the load. The microcontroller (ESP8266/ESP32) processes this data and calculates power consumption.The system is designed in such a way that it continuously monitors energy usage and transmits the data wirelessly. The overall design is simple, compact, and efficient, making it suitable for both domestic and industrial applications. The design ensures accurate measurement, easy installation, and user-friendly operation.

COMPONENTS USED :

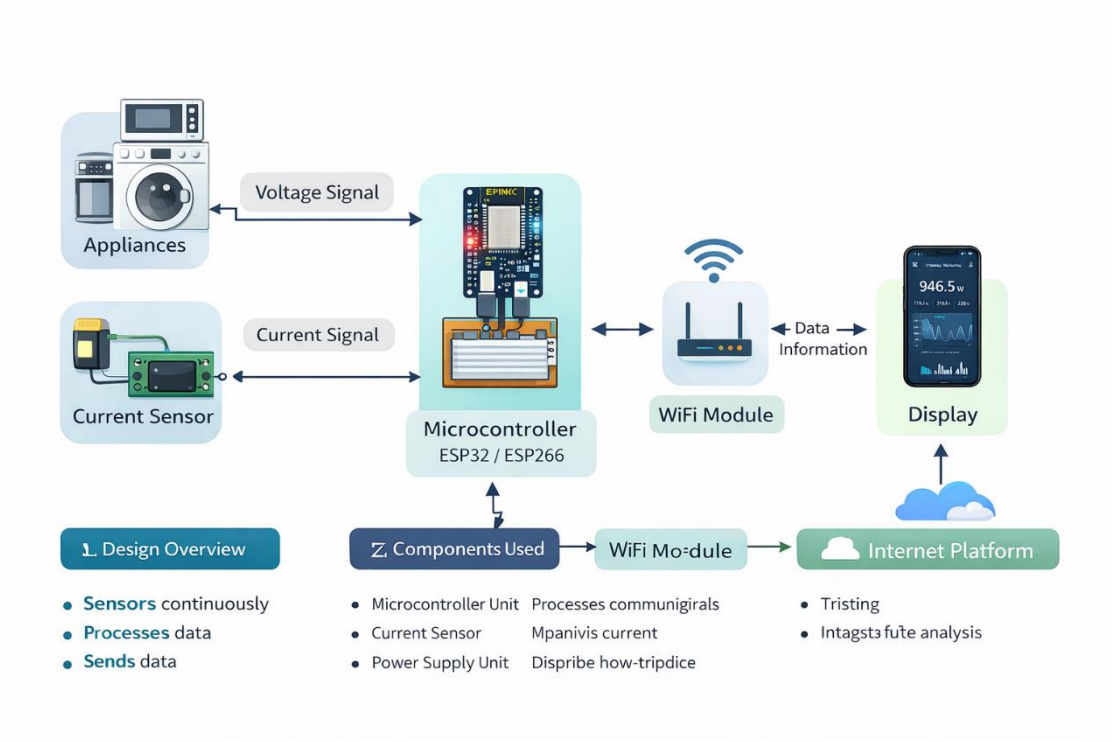
- **Microcontroller (ESP8266 / ESP32):** Processes data and handles communication
- **Voltage Sensor (ZMPT101B):** Measures voltage from the supply
- **Current Sensor (ACS712):** Measures current flowing through the load
- **Power Supply Unit:** Provides required voltage to the system
- **Connecting Wires:** Used for circuit connections
- **Breadboard / PCB:** Used for assembling the circuit
- **Load (Appliances):** Devices whose energy consumption is measured

FABRICATION PROCESS :

The fabrication process involves assembling and connecting all the components properly to build the system.First, the sensors are connected to the microcontroller according to the circuit design. The voltage sensor is connected in parallel with the load, while the current sensor is connected in series.

Proper connections are ensured to avoid errors. Next, the microcontroller is programmed using Arduino IDE or similar software. The code is uploaded to enable data processing and communication. After programming, the power supply is provided to the system.

Once the setup is completed, the system is tested to check whether the sensors are giving correct readings. Necessary adjustments and calibration are done to improve accuracy.



The above block diagram represents the overall structure of the IoT-based Smart Energy Monitoring System. It shows how different components are connected and how data flows through the system.

Initially, the electrical appliances act as the load, from which voltage and current values are measured using sensors. The voltage sensor and current sensor continuously monitor the electrical parameters and send the signals to the microcontroller.

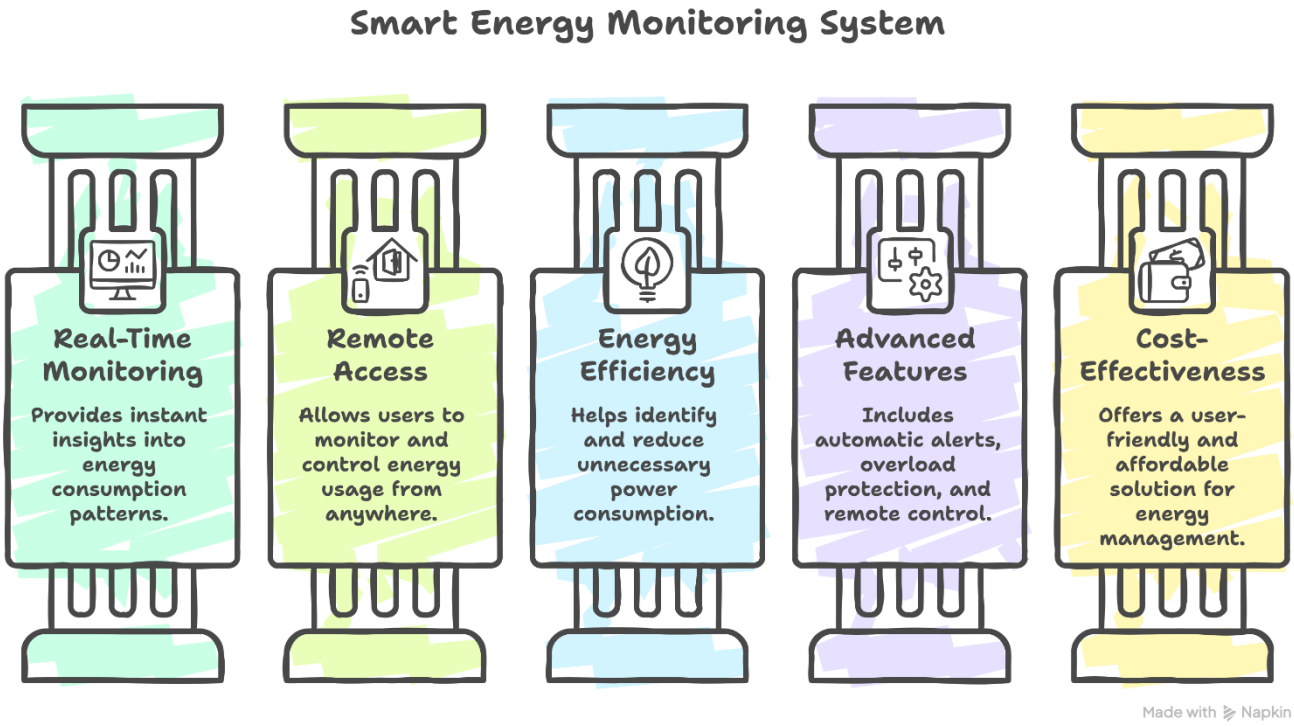
The microcontroller (ESP8266/ESP32) receives these signals and processes them to calculate power and energy consumption. It acts as the main control unit of the system.

After processing, the data is transmitted through the WiFi module. This enables wireless communication and allows the system to send data for monitoring purposes.

Finally, the processed information is displayed, helping the user to understand energy usage in real time. The system ensures continuous monitoring and efficient energy management.

CHAPTER 6

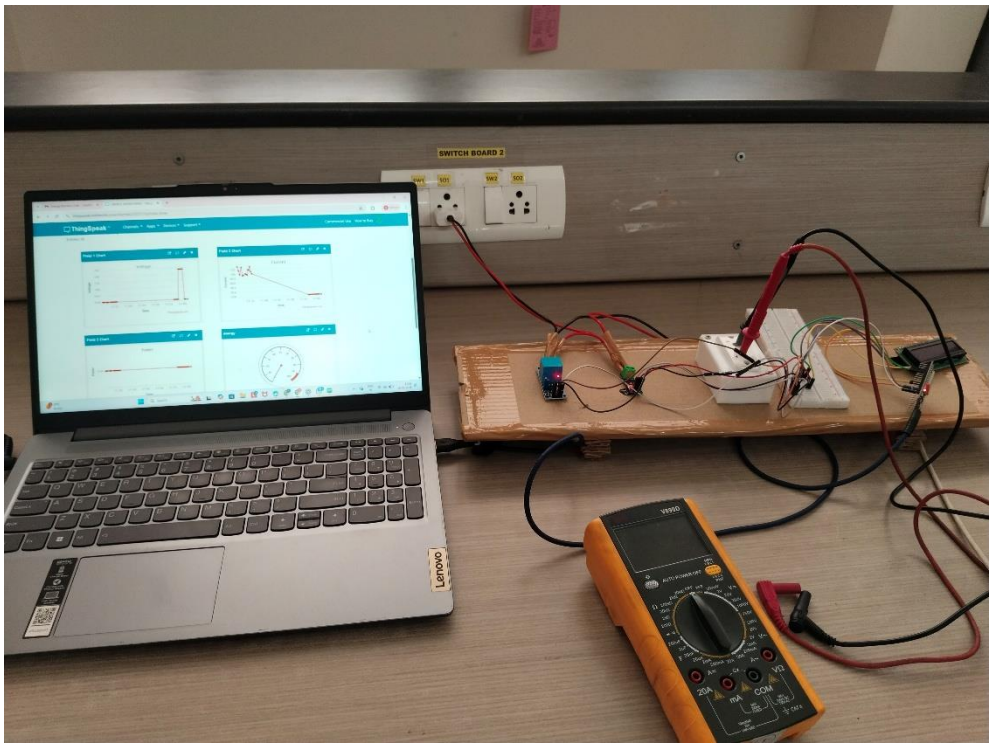
FLOWCHART

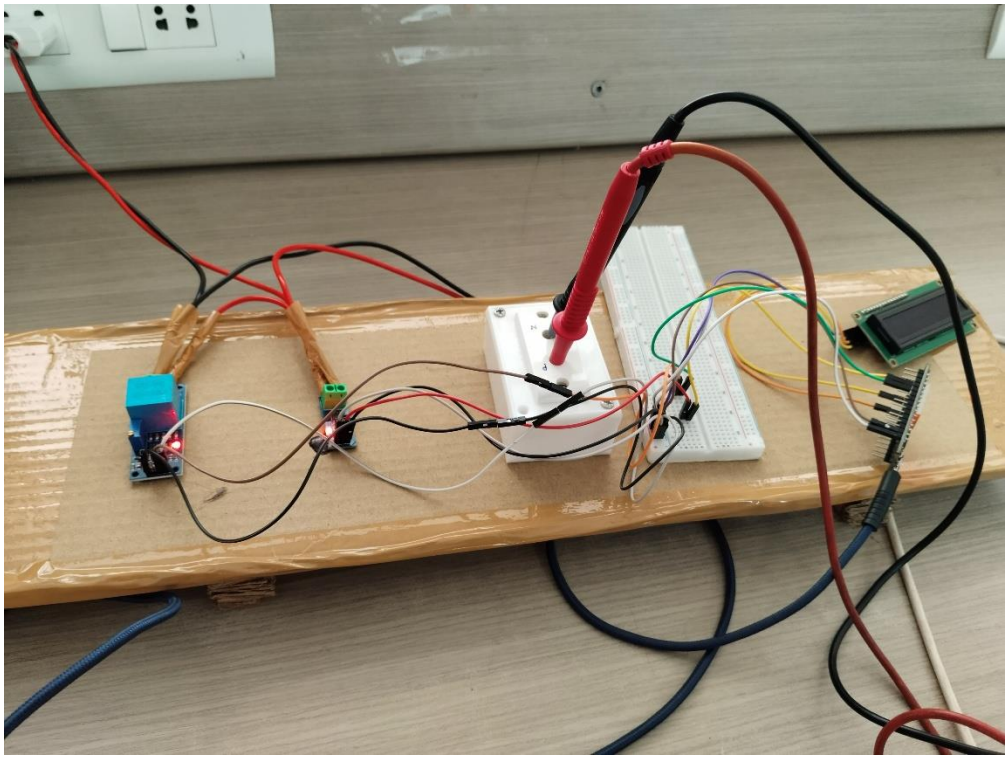


- **REAL-TIME MONITORING:** The system continuously tracks energy consumption and provides instant data to the user.
- **REMOTE ACCESS:** Users can monitor and control energy usage from anywhere using internet connectivity.
- **ENERGY EFFICIENCY:** Helps in identifying unnecessary power consumption and reduces energy wastage.
- **ADVANCED FEATURES:** The system supports features like automatic alerts, overload protection, and smart control of devices.
- **COST-EFFECTIVENESS:** It is an affordable and user-friendly solution that helps in reducing electricity bills

CHAPTER 8

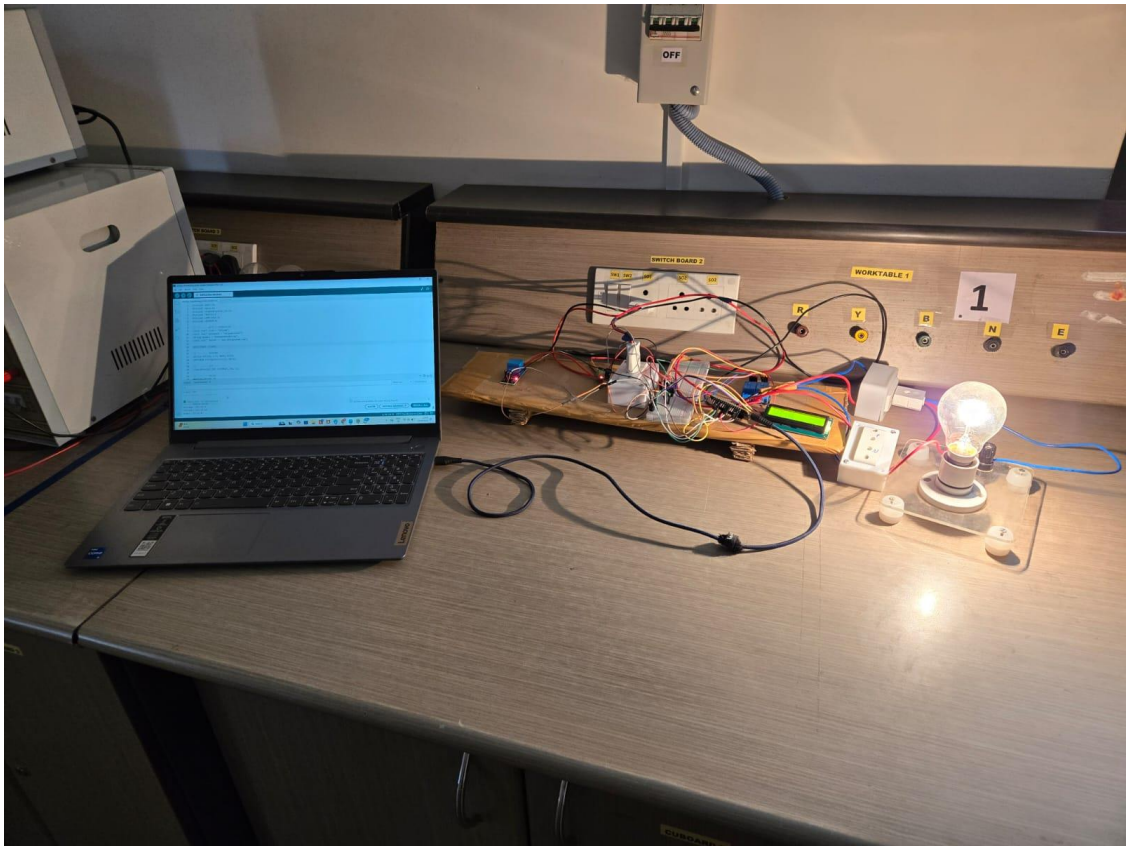
DESIGN OF SYSTEM





CHAPTER 7

RESULT



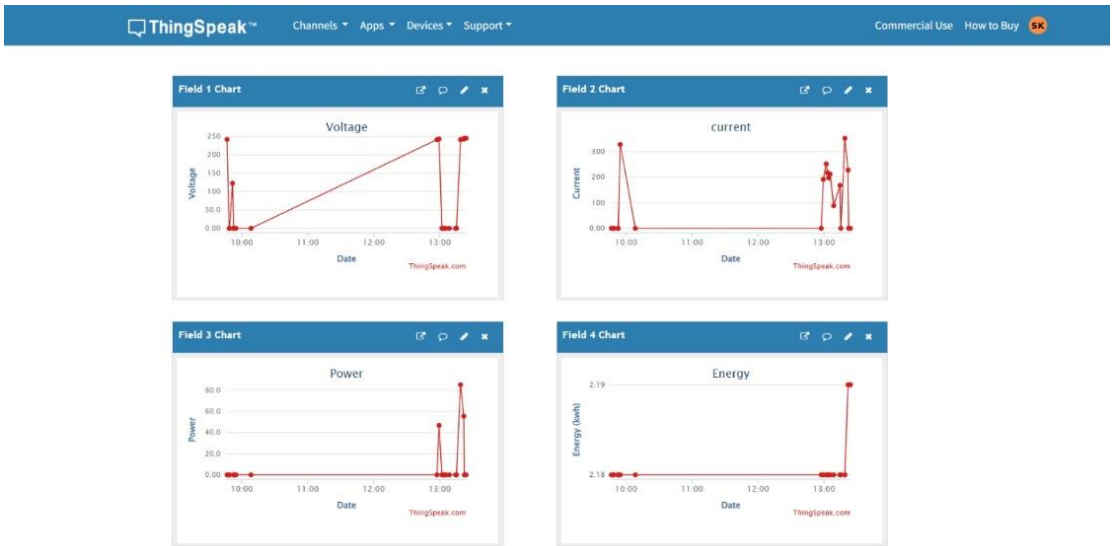
The IoT-based Smart Energy Monitoring System was successfully tested using a practical setup. The system includes a microcontroller circuit, sensors, a bulb as load, and a laptop for programming and monitoring.

During the experiment, the bulb was connected as an electrical load. When the bulb was switched ON, the system measured voltage and current using sensors. The microcontroller processed these values and calculated the power consumption.

The readings were updated in real time, showing that the system can continuously monitor energy usage. When the load condition changed (ON/OFF), the system responded quickly and updated the values correctly.

The hardware setup worked properly without major errors. All components were connected correctly, and the system showed stable performance during testing.

DATA ANALYSIS :



The above graphs show the real-time data of voltage, current, power, energy, and load status obtained from the IoT-based Smart Energy Monitoring System

VOLTAGE GRAPH :

From the voltage graph, it is observed that the voltage remains mostly stable around the supply value, with slight variations during operation. This indicates that the system is receiving proper input voltage.

CURRENT GRAPH :

The current graph shows that current increases when the load (bulb) is switched ON and drops to zero when it is OFF. This clearly indicates the working condition of the load.

POWER GRAPH :

The power graph is directly related to voltage and current. When the load is ON, the power value increases, and when the load is OFF, the power becomes zero. This shows correct calculation of power by the system.

ENERGY GRAPH :

The energy graph shows a gradual increase over time, which means energy consumption is accumulating continuously when the load is in use.

LOAD 1 AND LOAD 2:

The Load 1 and Load 2 status graphs indicate the ON/OFF condition of the appliances. A value of “1” represents ON state, and “0” represents OFF state. These graphs confirm that the system can monitor multiple loads effectively.

CHAPTER 9
CONCLUSION

The IoT-based Smart Energy Meter and Load Control system was successfully designed and implemented to monitor and control electrical energy usage in real time. The system effectively measures voltage, current, power, and energy consumption using sensors, and processes the data using a microcontroller.

In addition to monitoring, the system also provides load control functionality, allowing electrical appliances to be switched ON and OFF based on user requirements. This helps in reducing unnecessary power consumption and improves overall energy efficiency.

The system provides accurate and continuous data, enabling users to understand their energy usage patterns and take necessary actions to save electricity. It reduces manual effort, improves convenience, and enhances safety through better control of electrical loads.

The developed system reduces the need for manual meter reading and provides better control over electrical devices. It helps users to identify high power-consuming appliances and take corrective actions to minimize energy wastage. The continuous monitoring and control features enhance user convenience and improve awareness about energy usage.

Furthermore, the system supports future enhancements such as automation, smart alerts, and integration with advanced technologies. It contributes to energy conservation, reduces electricity costs, and promotes sustainable usage of resources.

Overall, the project demonstrates the effective use of IoT technology for smart energy management and automation. It is a cost-effective, reliable, and user-friendly solution that can be used in homes, industries, and commercial applications. The system plays an important role in energy conservation and supports the development of smart and sustainable environments.

CHAPTER 10

REFERENCE

1. **INTERNET OF THINGS: A HANDS-ON APPROACH"**

   Authors: Arshdeep Bahga, Vijay Madisetti

   Overview: This book provides a comprehensive introduction to IoT concepts, including sensors, communication protocols, and real-time applications, which are essential for developing smart energy monitoring systems.

2. **IOT BASED SMART ENERGY MONITORING SYSTEM"**

   Authors: Various Authors (IEEE Research Papers)

   Overview: This paper explains the design and implementation of IoT-based systems for real-time energy monitoring, focusing on improving efficiency and reducing power wastage

3. **"SMART ENERGY MONITORING SYSTEM USING IOT"**

   Authors: S. M. Ali, R. Gupta, P. Sharma

   Overview: This research work presents an energy monitoring system that uses IoT technology to track electricity usage and provide remote access to users for better energy management.

4. **"IOT-BASED ENERGY MANAGEMENT SYSTEM FOR SMART HOMES"**

   Authors: A. Kumar, S. Singh

   Overview: This study discusses the integration of IoT in smart homes to monitor and control energy usage, improving efficiency and reducing electricity consumption.

5. **ESP8266/ESP32 TECHNICAL REFERENCE MANUAL**

   Authors: Espressif Systems

   Overview: This documentation provides detailed information about the microcontroller used in IoT applications, including WiFi communication and data processing.

6. **ACS712 CURRENT SENSOR DATASHEET**

   Authors: Allegro Microsystems

   Overview: This datasheet explains the working and specifications of the current sensor used to measure electrical current in energy monitoring system.